

(a) A GPU Read Copies

private writable mapping



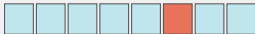
GPU read



512 pages (2 MiB)
become private copies

(b) A Flush per Page

address space of the process



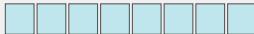
copy-on-write break

SMMU invalidation
for the whole process

about 9 us per page;
4.3 to 9.0 times slower

(c) fork Leaves a Debt

memory of the parent



fork



every page write-protected

the next GPU write pays
2.7 s per GiB